

Power Electronics Laboratory

Session 3: DC-AC converter 1

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Objectives

1. Study basics topologies of DC/AC power converters.
2. Study the impact of the modulation scheme on the H-Bridge converter.
3. Generation of a sinusoidal output voltage (Without control).

General instructions

1. Introduction is evaluated with **0.5** points maximum.
2. Conclusion is evaluated with **0.5** points maximum.
3. For every **3** figure format and order error, **0.1** points will be discounted.
4. All figures and equations must be referenced.
5. In many tasks, it is not indicated what variables must be plotted. The student must be able to decide what variables to plot, and how to plot them.

Part 1

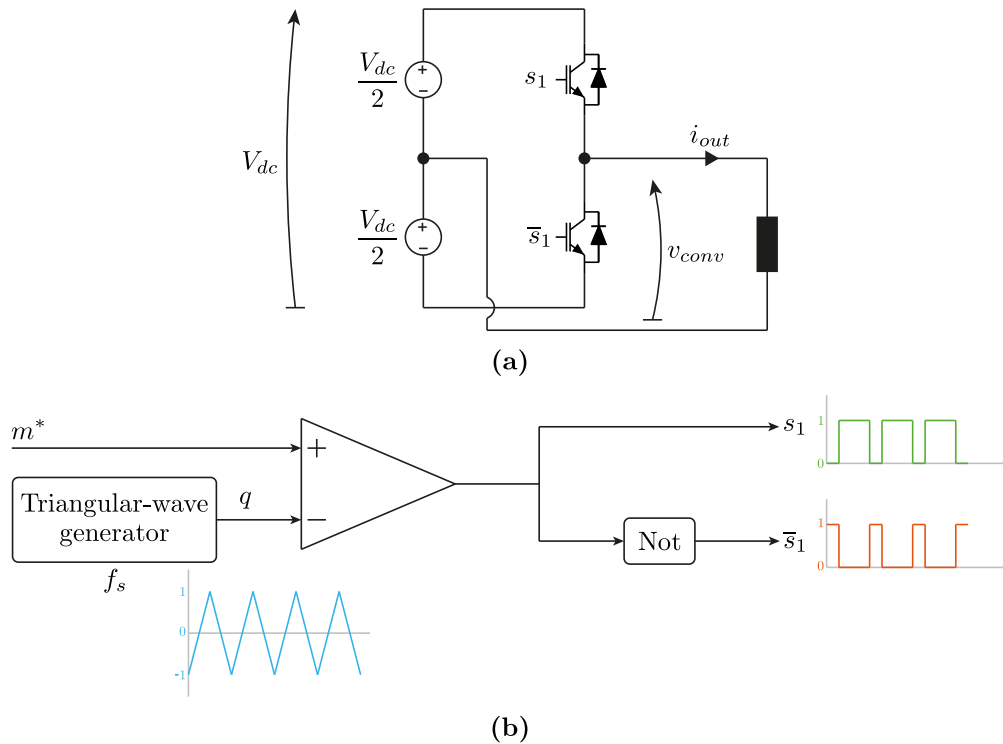


Figure 1: Half-Bridge Converter. (a) Topology. (b) Modulation scheme.

Tasks

Table 1: System parameters.

Parameter	Value
V_{dc}	500V
Switching frequency	10 kHz
Series RL load	$L = 10\text{mH} - R = 10$

s_1 (On - OFF)	v_{conv}

1. For the Half-Bridge topology of Fig. 1, write all switching states possible, and the output voltages of each state.
2. Implement the Half-Bridge topology of Fig. 1(a) with the modulation diagram of Fig. 1 For the Half-Bridge topology of Fig. 1(b).
3. Choose three different modulation indexes. For each case calculate the fundamental component (50 Hz) of the output voltage of the converter, the fundamental component of the output current, and the active power delivered. **Justify your calculations.** Verify it using the simulation model.
4. Measure the active power on the input side and compare it with the output active power. **Comment.**
5. Once the steady state is achieved, obtain the frequency spectrum of the power converter voltage and output current. The base frequency is 50 Hz, and the maximum frequency is 50 kHz. Identify the switching frequency and **comment.**

Part 2

Tasks

1. For the H-Bridge topology of Fig. 2, identify and write all the possible switching states, and the output voltages associated with each state.

s_1 (On - OFF)	s_2 (On - OFF)	v_{conv}

2. Implement the circuit of Fig. 2(a), with the bipolar modulation shown in Fig. 2(b). Identify which commutation states are being utilized.
3. Implement the circuit of Fig. 2(a), with the unipolar modulation shown in Fig. 2(c). Identify which commutation states are being utilized.
4. Choose three different modulation indexes. For each case calculate the fundamental component of the output voltage of the converter, the output current, and the active power delivered. **Justify your calculations.** Verify it using the simulation model.
5. Measure the active power on the input side and compare it with the output active power. **Comment.**
6. With Bipolar and Unipolar modulation, once the steady state is achieved, obtain the frequency spectrum of the power converter voltage and output current. The base frequency is 50 Hz, and the maximum frequency is 50 kHz. Identify the switching frequency and **comment.**

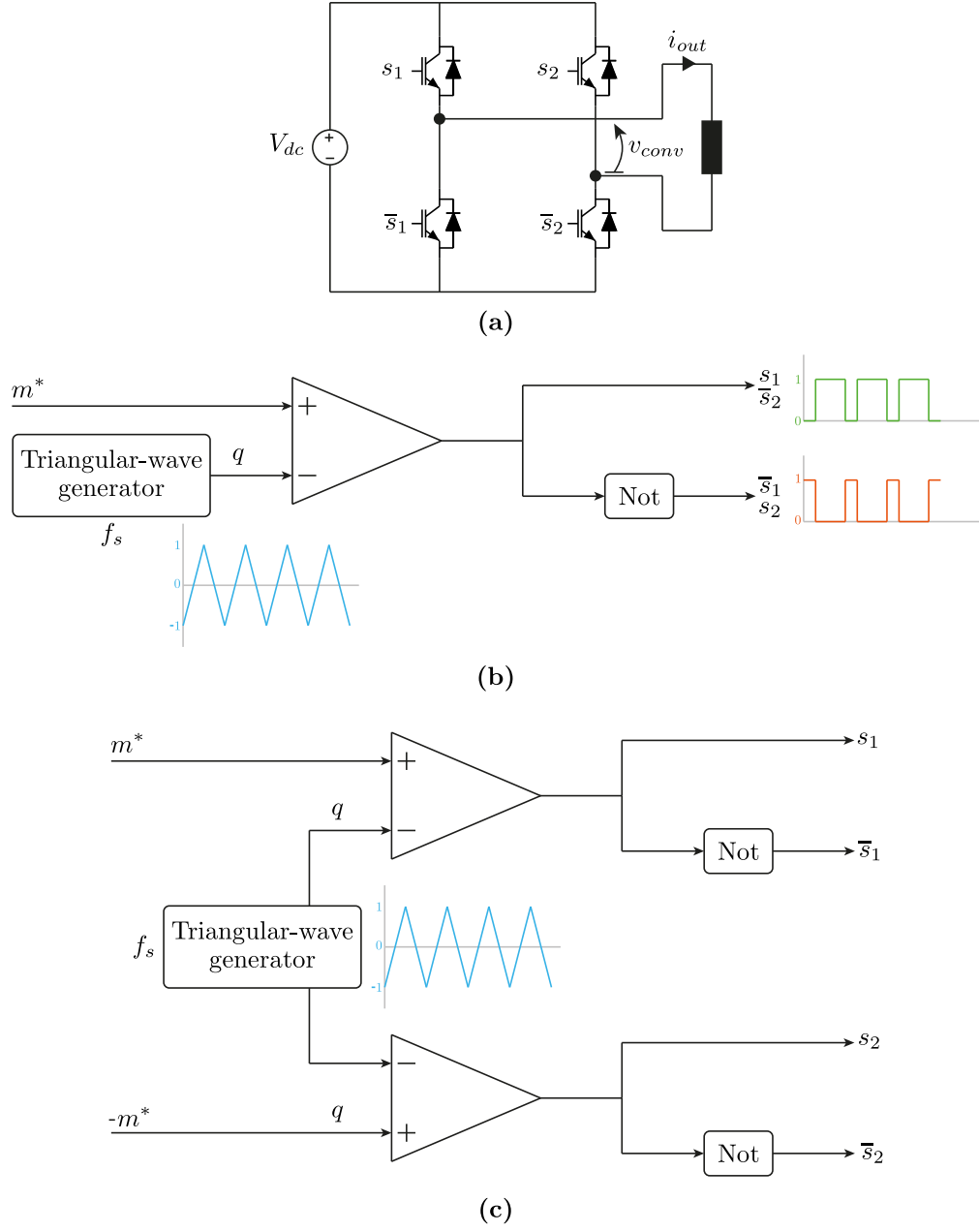


Figure 2: H-Bridge Converter. (a) Topology. (b) Bipolar modulation scheme. (c) Unipolar modulation scheme.